

Praca magisterska

Projekt przekształtnika Bridgeless Totem-Pole
PFC z tranzystorami z azotku galu oraz
cyfrowym sterowaniem opartym na
mikrokontrolerze STM32H7

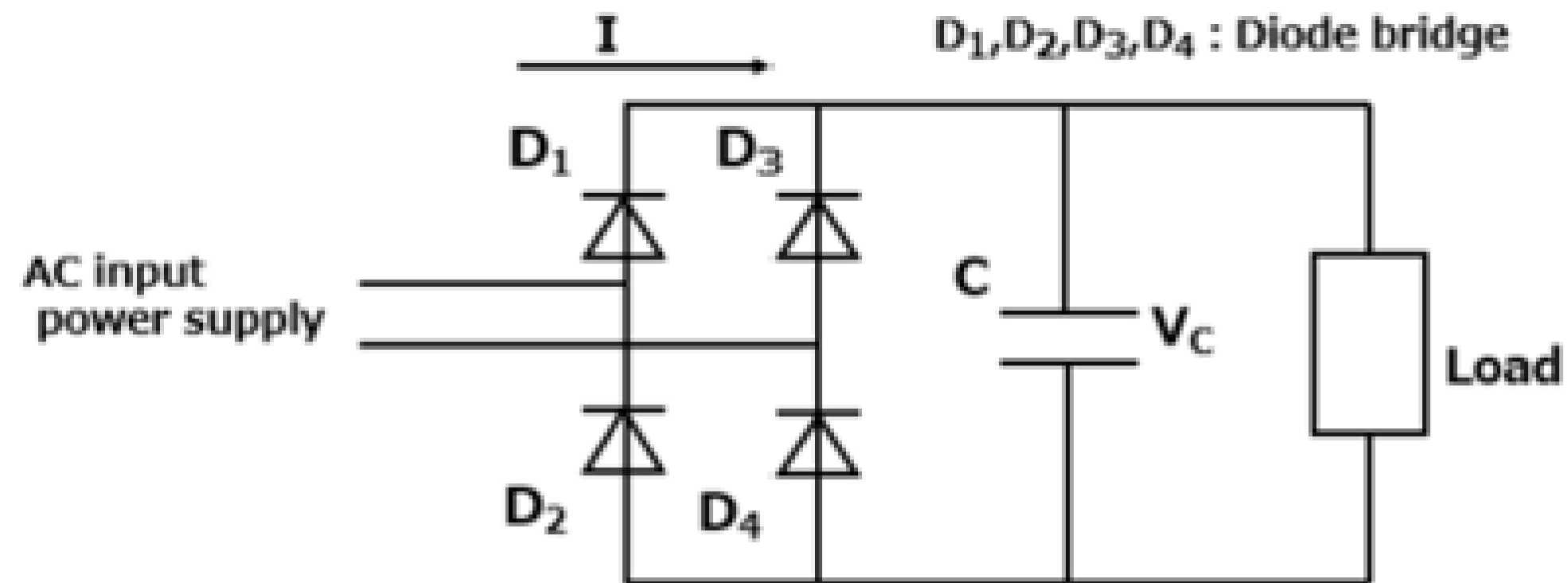
Błażej Skrzypkowski

Promotor: *dr inż. Rafał Kopacz*

**Politechnika
Warszawska**

Warszawa, 14 kwietnia 2026 r.





$$PF = PF_{DISPLACEMENT} \times PF_{DISTORTION} = \cos(\theta_V - \theta_I) \times \sqrt{\frac{1}{1 + THD^2}}$$

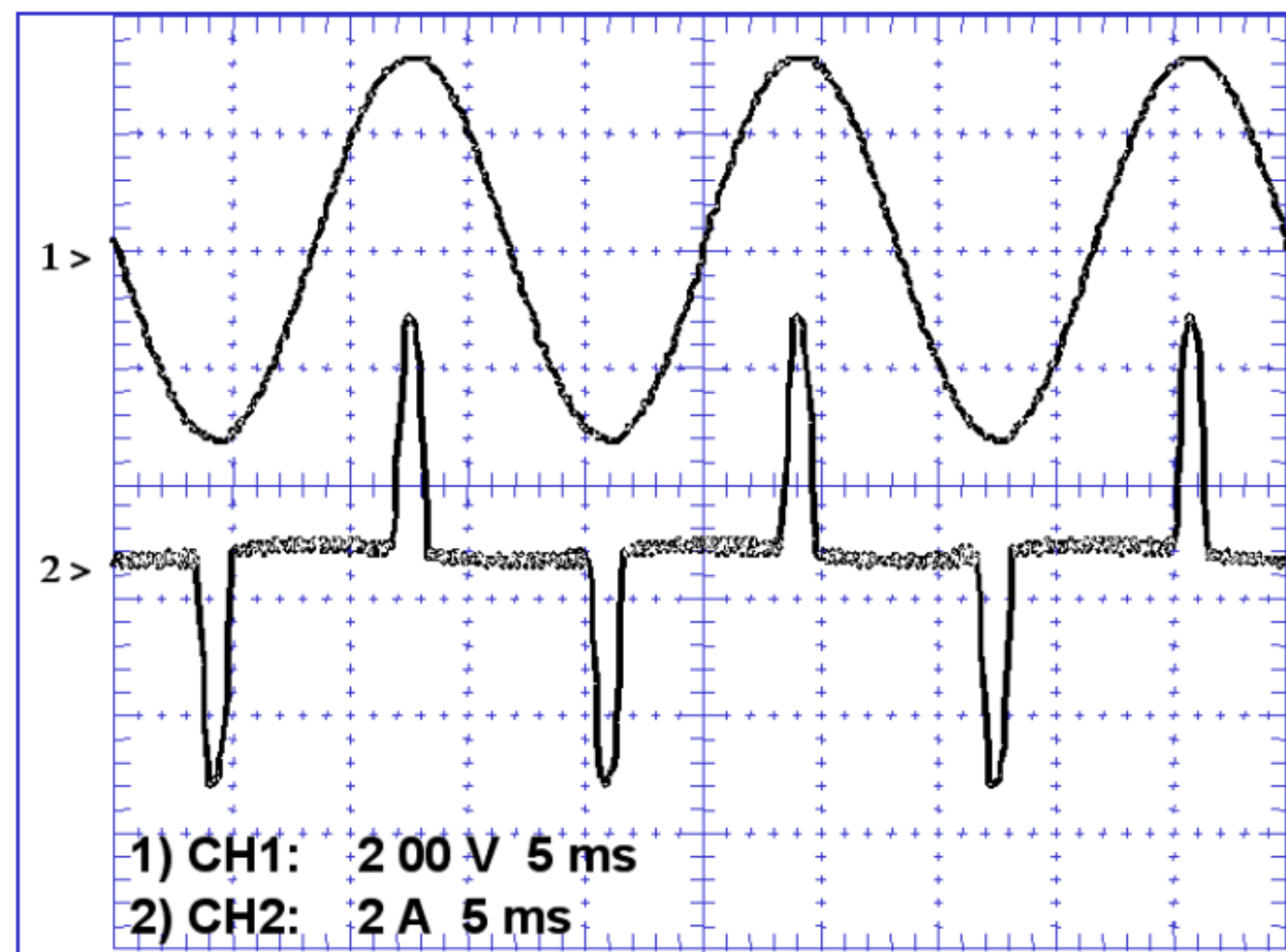
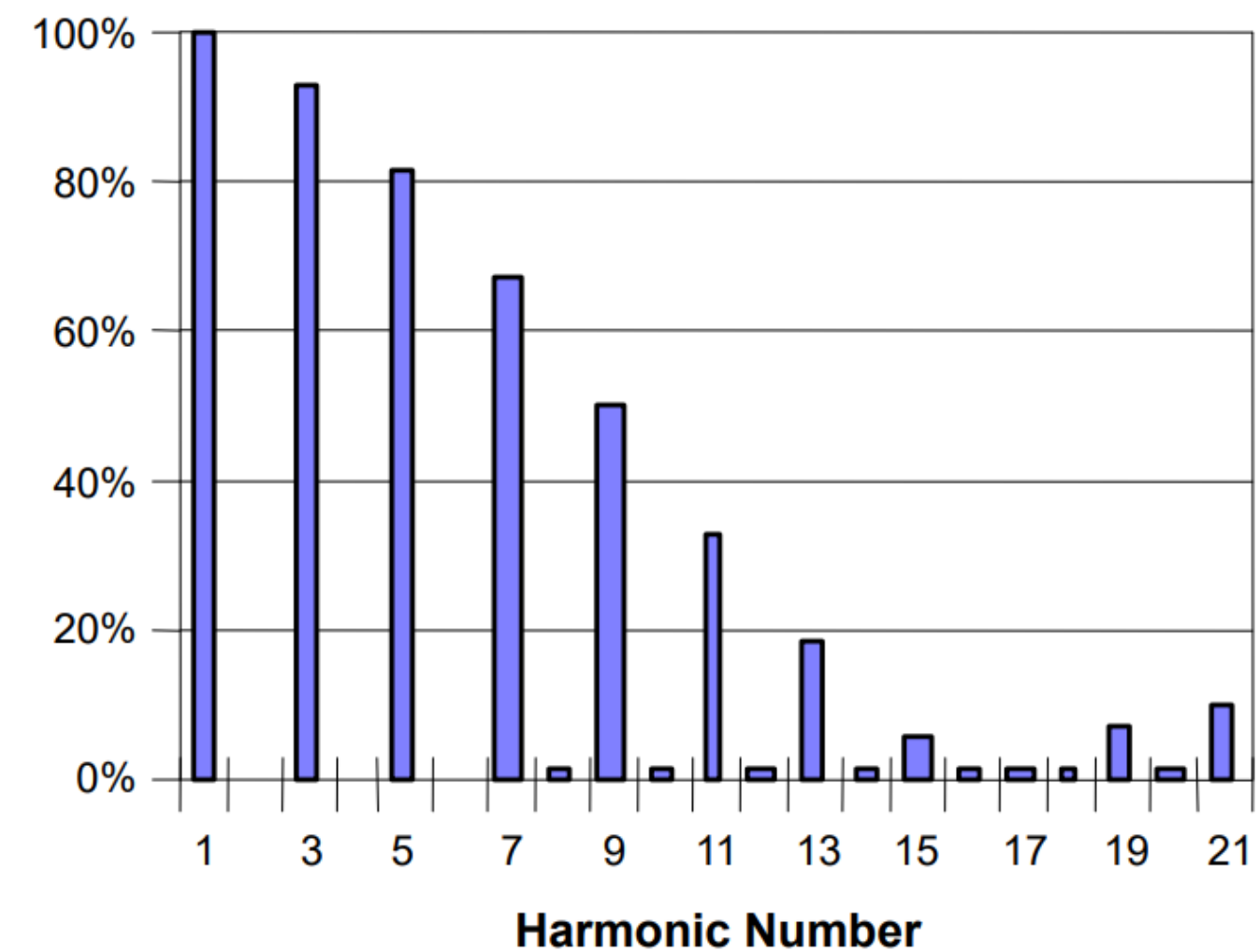
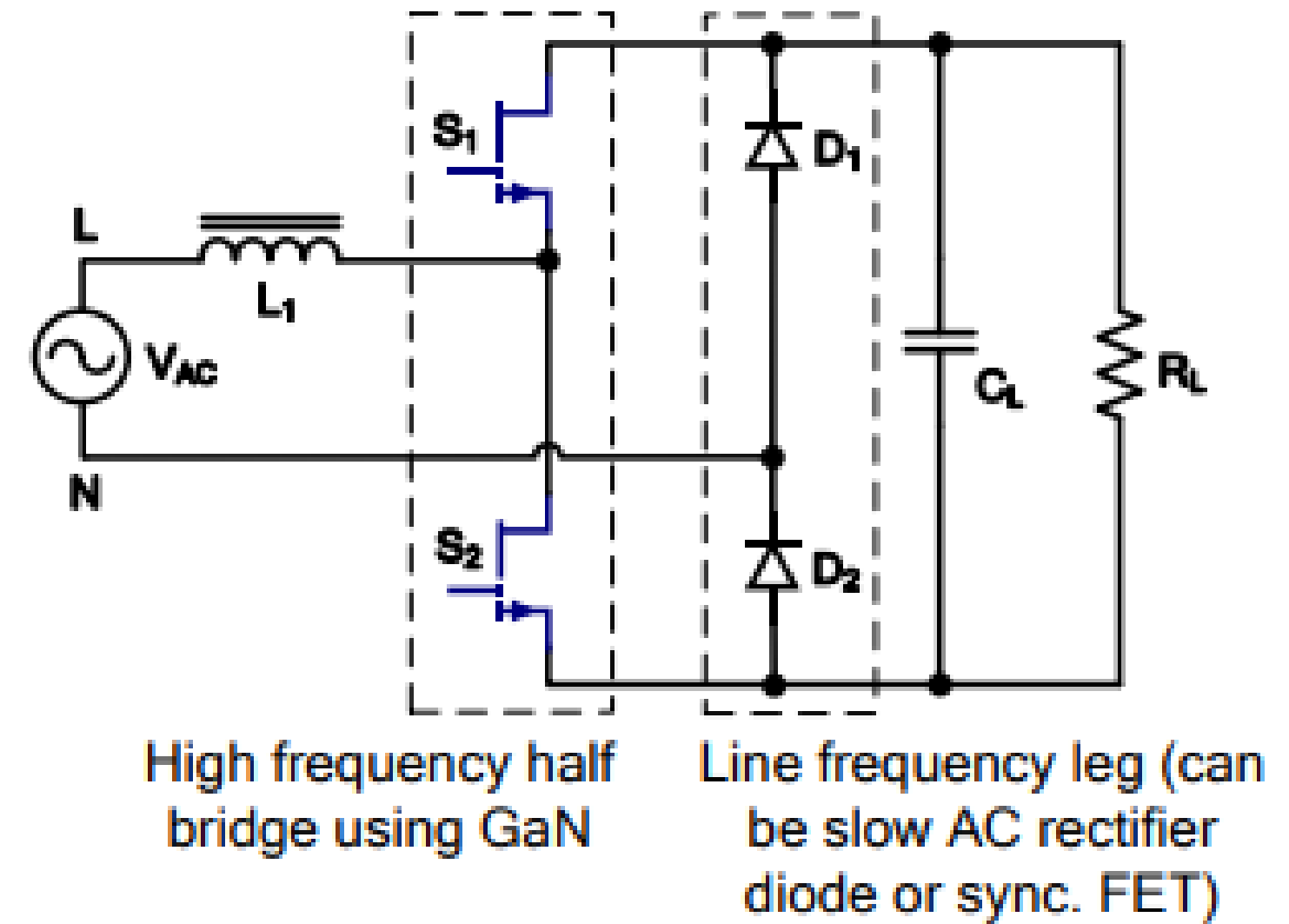
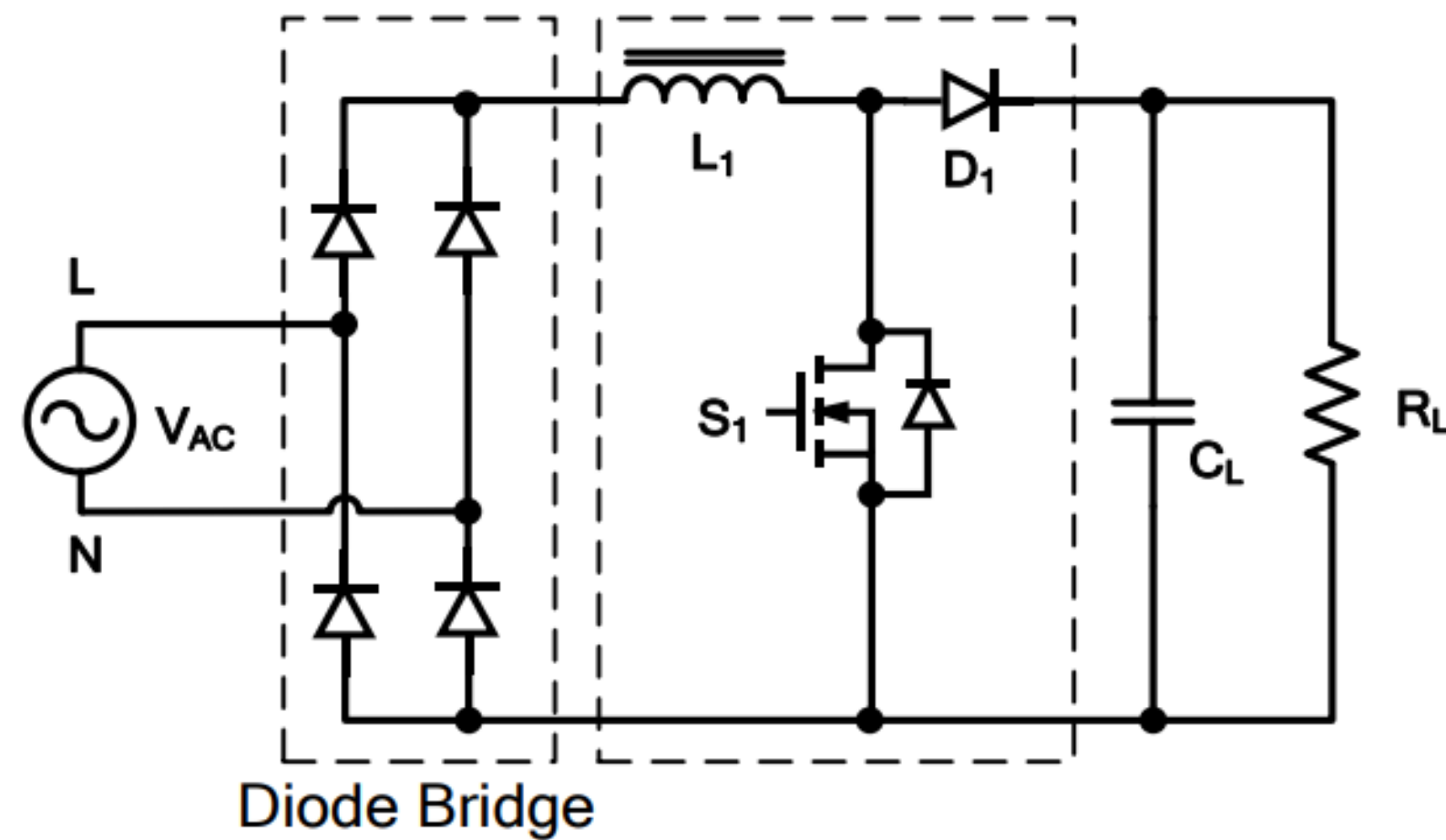
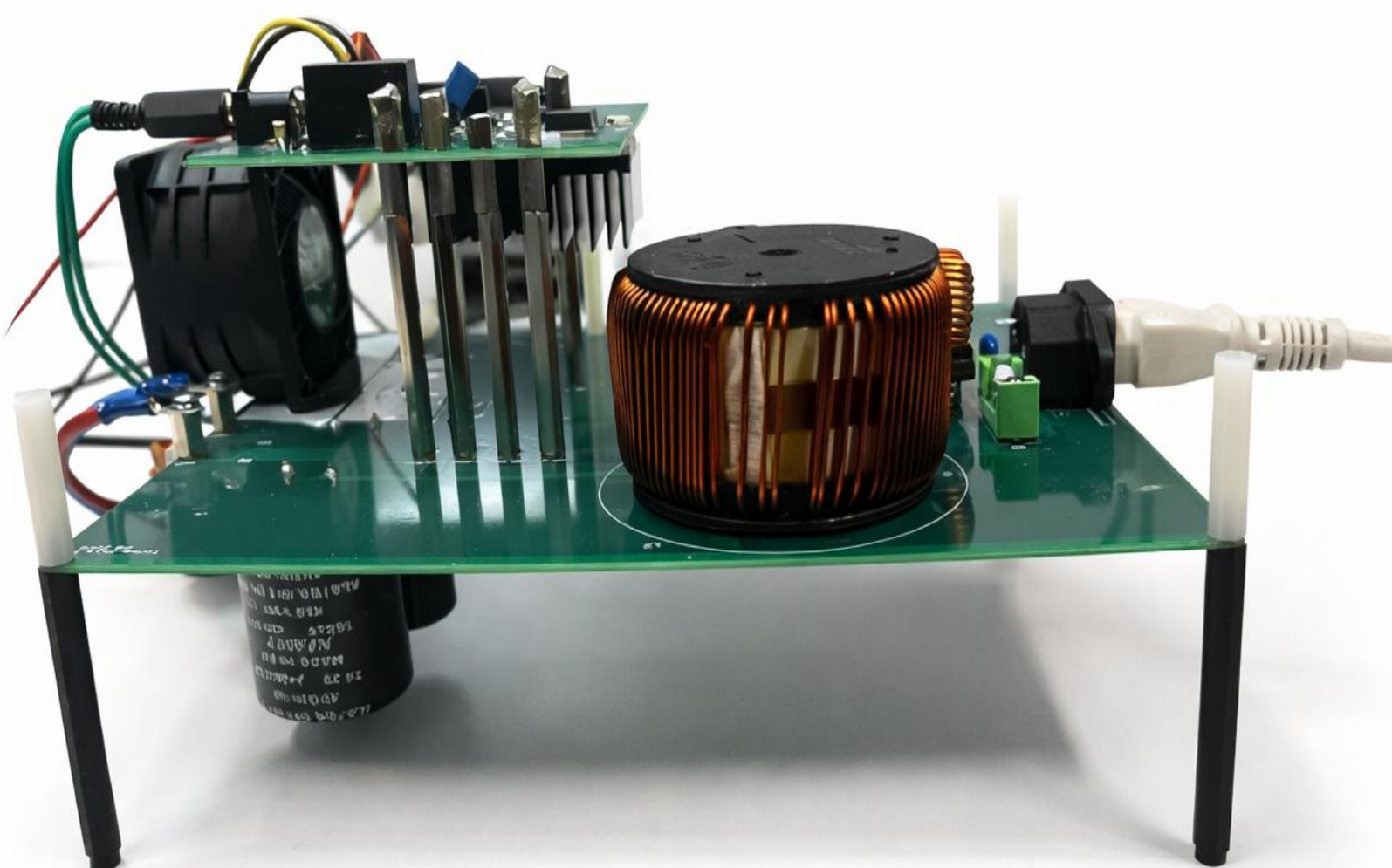
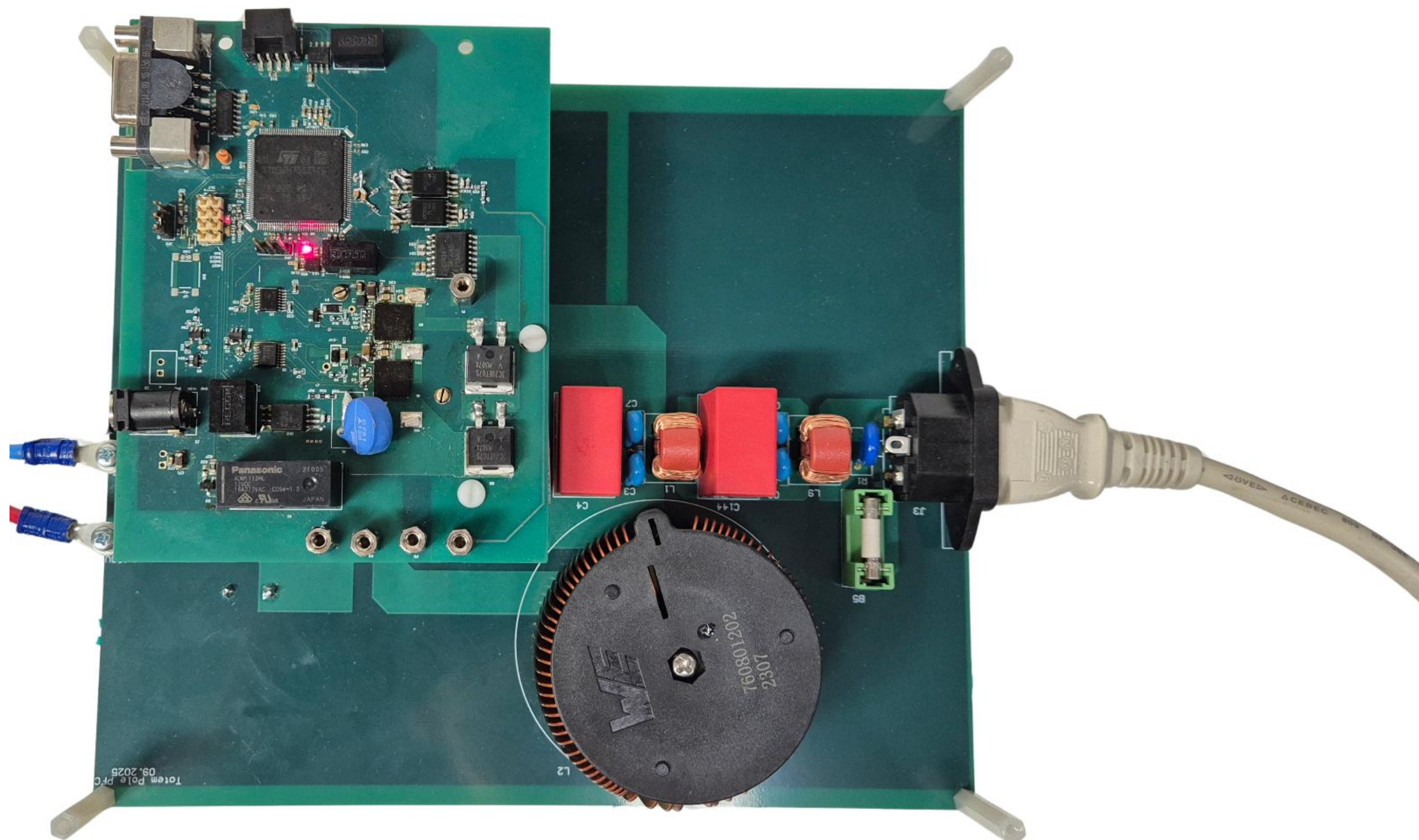


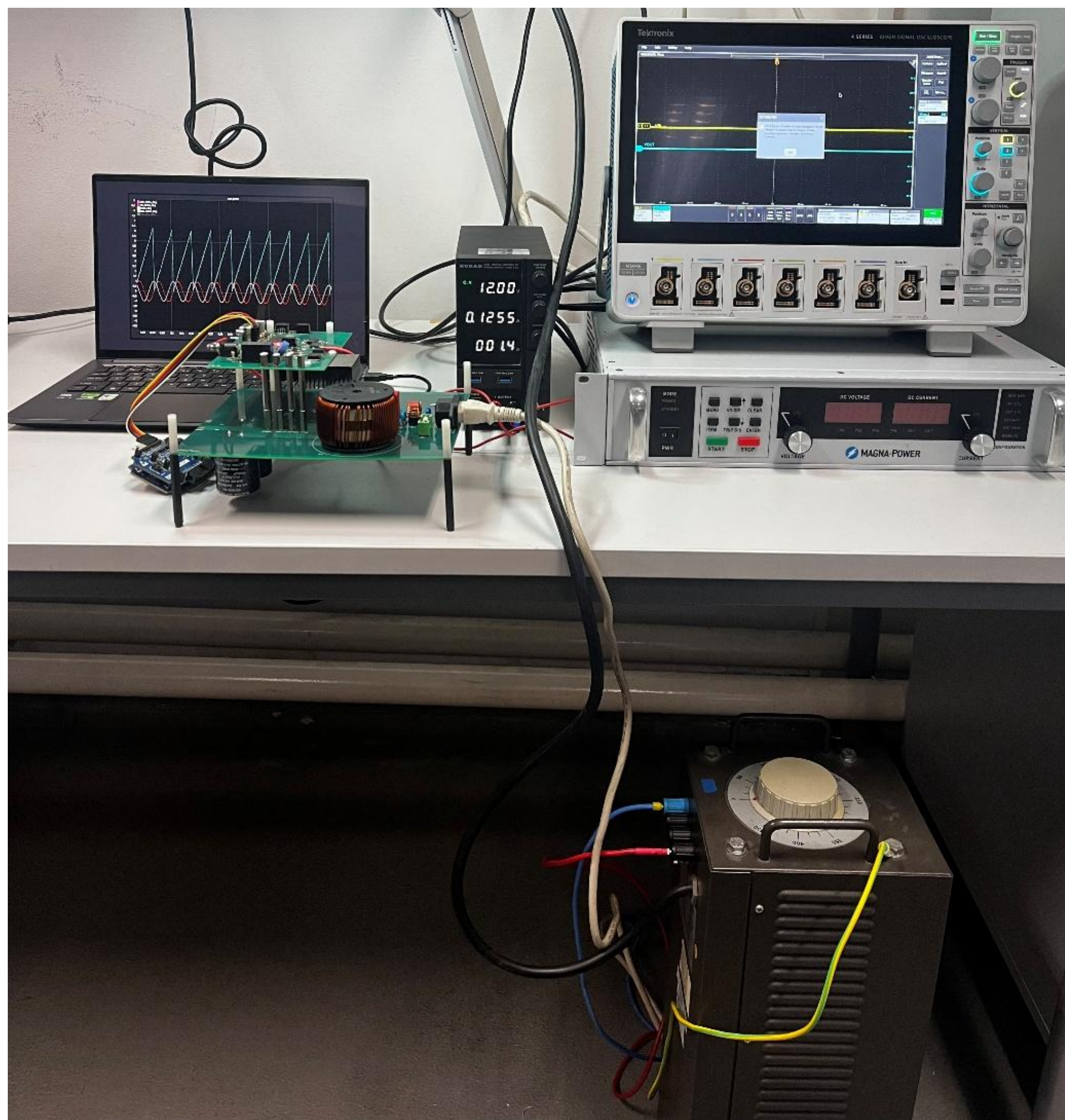
Figure 1-1. Input Characteristics of a Typical Switched-Mode Power Supply without PFC

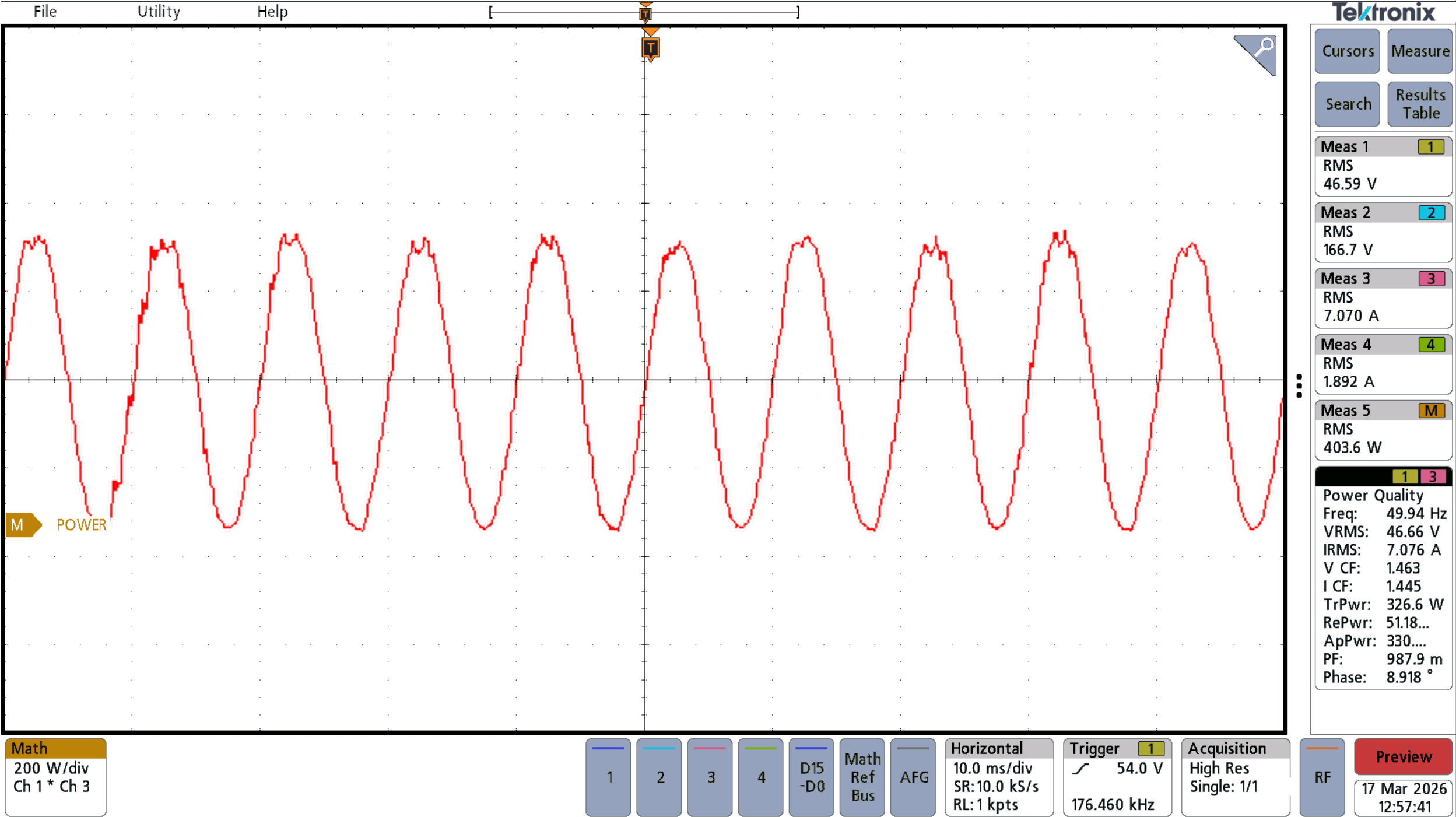
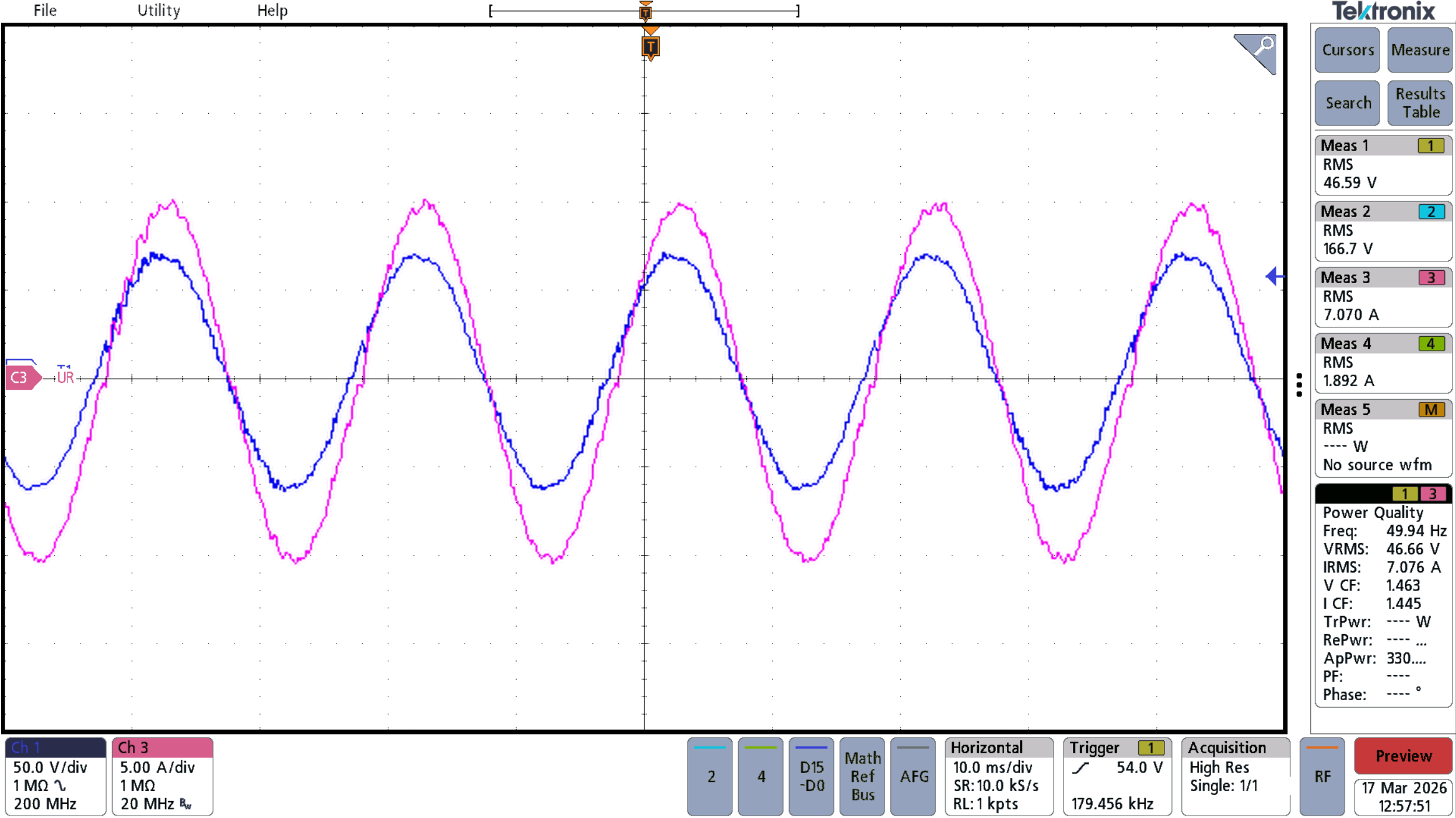




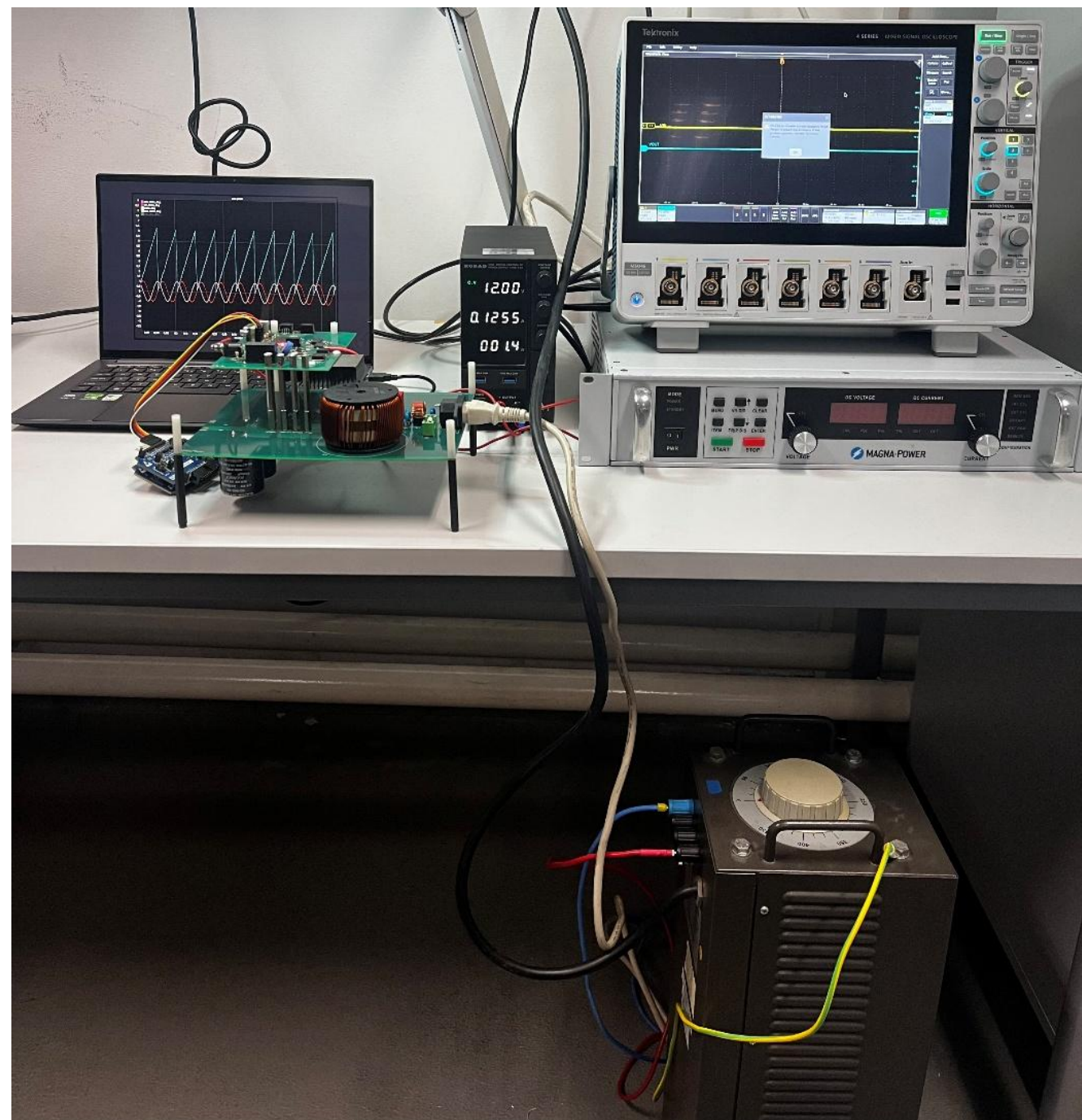
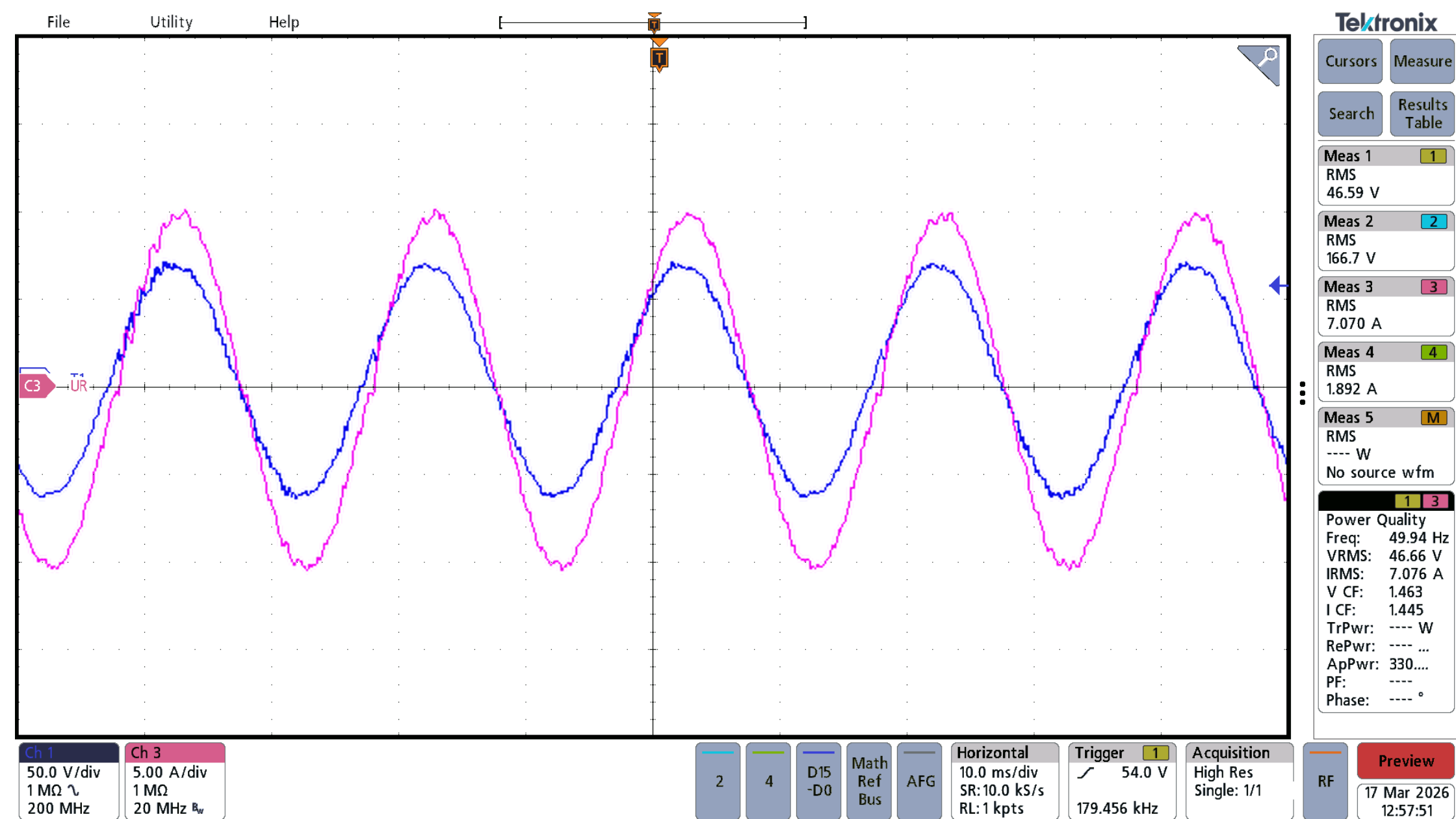
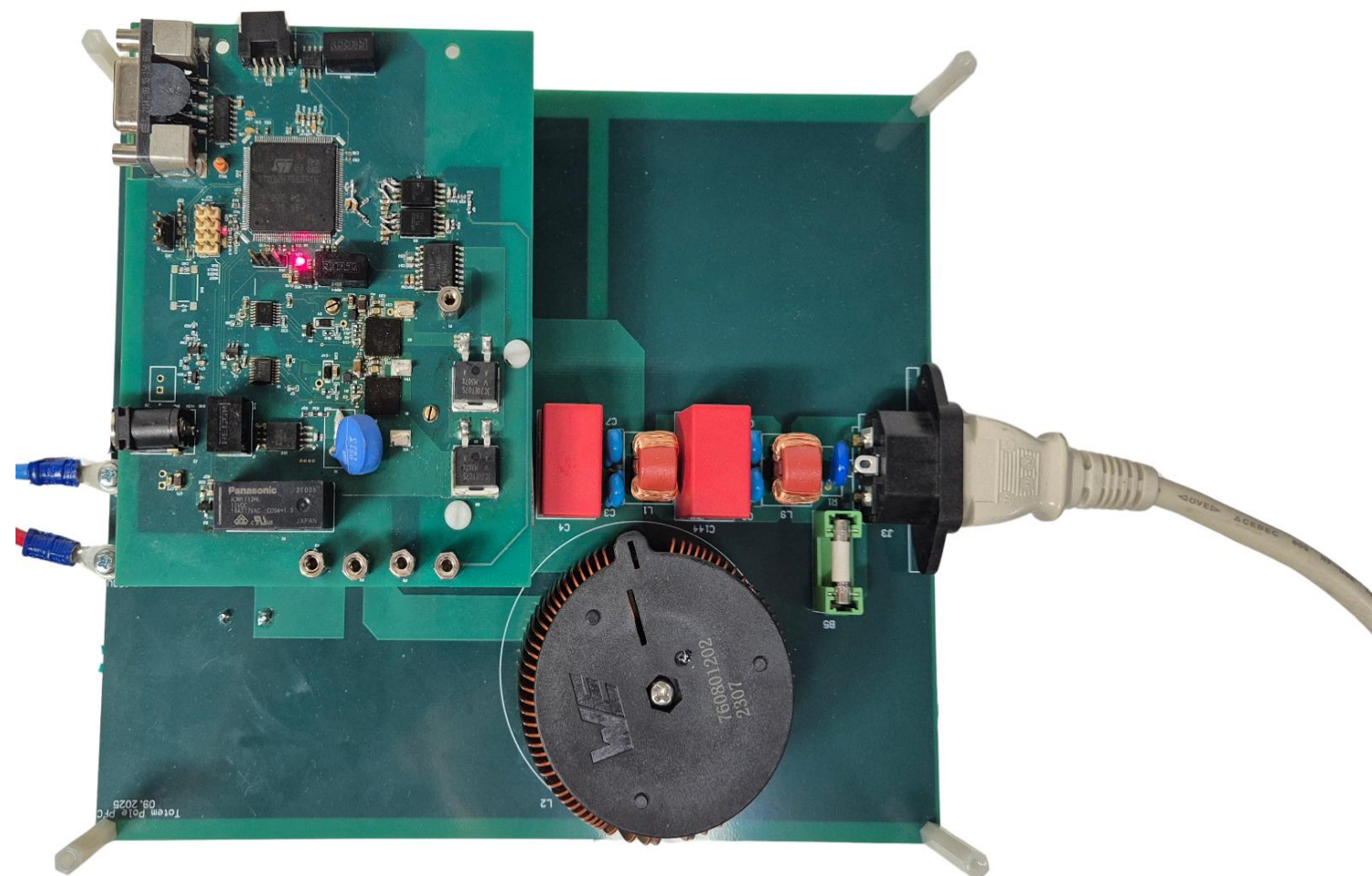
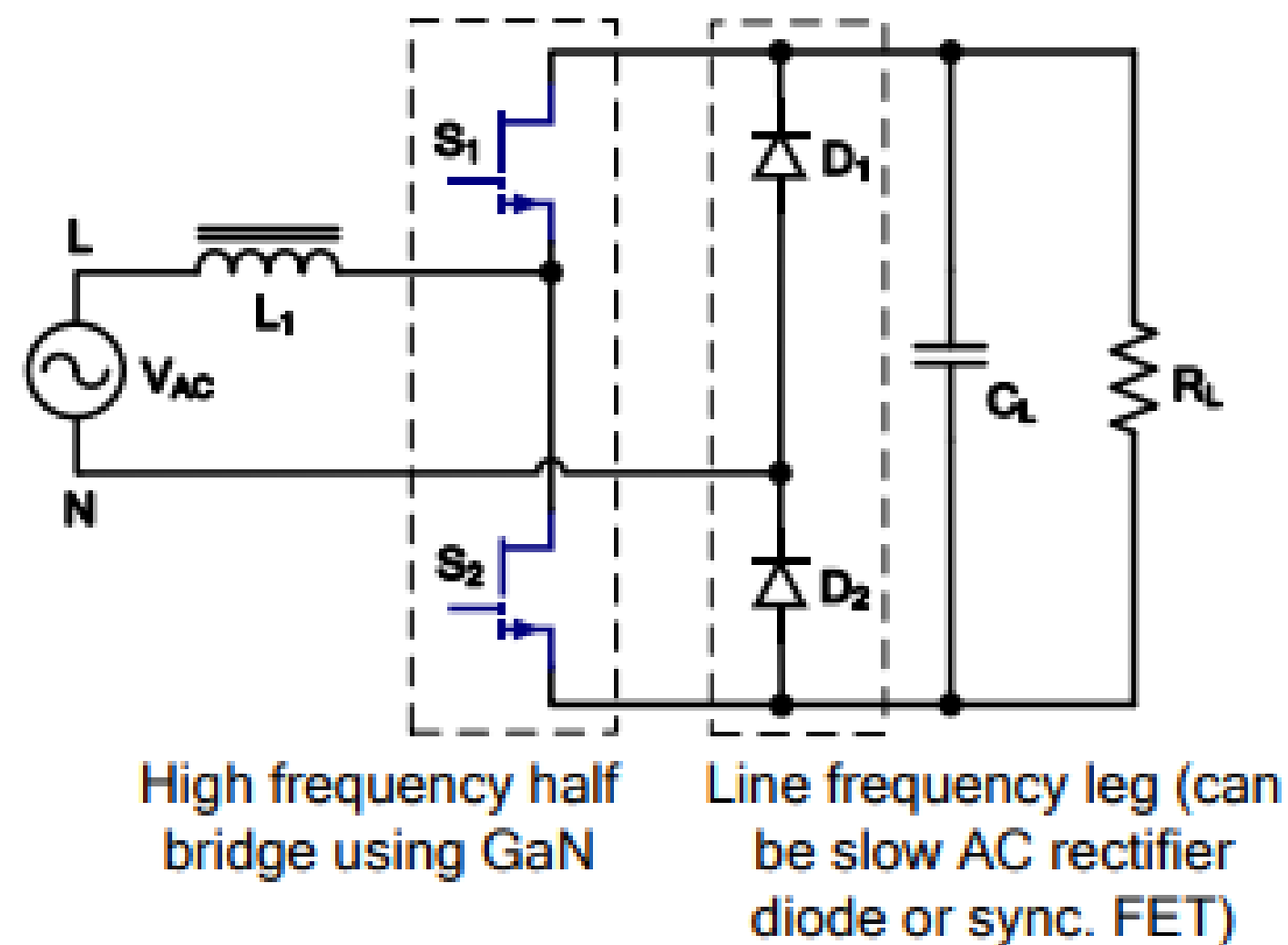
	Si CoolMOS CFD7 w/ Fast Body Diode	GaN HEMT	
	IPW65R029CFD7	GS66516B	Unit
$R_{DS(ON)}$ Typ.	24	25	mΩ
Q_{rr}	1600	0	nC
Q_{OSS} @ $V_{DS}=400V$	1012	134	nC
Turn-on loss due to $Q_{RR}/Q_{OSS}@F_{SW}=65kHz$	68.7	2.5	W







Design and Implementation of a Bridgeless Totem-Pole PFC Converter with GaN Transistors



Dziękuję za uwagę

**Politechnika
Warszawska**

Błażej Skrzypkowski

